Bachelor Project Future Planet Studies

Quantifying Mexico City's urban water supply security: an integrated and critical analysis



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Terminology and abbreviations

A:	social capital
CPA:	Capital Portfolio Approach
D_W:	volumetric water demand
F:	financial capital
g:	supply gaps bridged by storing and rationing water use at household level
h:	the fraction of households connected to the public water supply infrastructure
I:	institutional capital
N:	natural capital
P:	physical capital
SACMEX	Sistema de Aguas de la Ciudad de México
SETS:	social-ecological-technological systems
S_W:	the fraction of water supply that is delivered to the households
UWA:	urban water availability
UWSS:	urban water supply security
W_{extra}:	additional water resources accessed by the community through informal sources
$W_{Leakage}$:	the amount of leakage [m^3y^{-1}]
WS:	Water Stress

1. Abstract

The Capital Portfolio Approach (CPA) is a framework that analyses the urban water supply security (UWSS) on a case study basis. I apply this framework to investigate Mexico City, a highly populated area situated in a valley and constructed on a drained lakebed. The region is pressured by, amongst others, land subsidence, earthquakes, water leakage, low-quality water, and distribution problems. All these factors negatively impact the UWSS. This research conducts an analysis and quantification of the five 'capitals' presented and operationalised in the CPA. The aim is to formulate an integrated overview of these capitals, including the weaknesses and strengths of the system. The outcome of this research is that Mexico City lacks the physical, institutional, and financial capitals to efficiently contribute to UWSS. Therefore, the adaptability of the inhabitants, the social capital, plays larger role than potentially necessary. The current trajectory of Mexico City's water supply system is likely to intensify the hazards to which the region is susceptible. This will aggravate the hazardous effects on the socioeconomic situation of Mexico City on every level of organisation. The most important leverage points in the Mexico City water system therefore are improving the hydraulic infrastructure to decrease high leakage rates, improving the socioeconomic stability, which is driven by trust in government, income equality and managerial accountability. Focusing on and strengthening these aspects will increase the overall system's functioning. This interdisciplinary research can form the basis from which policy makers and interest groups can develop an educated response to adapt and mitigate to the challenges Mexico City faces regarding UWSS.

2. Introduction

Over the past decades, urban areas have become increasingly more popular to move to and to settle in. Nowadays, over 55% of the world's population lives in urban areas and this percentage of urbanisation is expected to grow to 68% in 2050 (UNDESA, 2018a). Metropolitan areas are under more pressure to provide their inhabitants with the basic human needs and maintain sufficient amounts of natural resources. These pressures, and potential problems that arise from them, will increase with the additional one billion world citizens projected to move to urbanized regions in the upcoming three decades.

Access to safe (drinking) water is an internationally recognised human right, thus should be available to all world citizens (OHCHR, 2010). To obtain or safeguard this human right in urbanised areas, there is a necessity to understand and quantify the interacting components involved in the provisioning of water to the inhabitants. The interdisciplinary 'Capital Portfolio Approach' (CPA) is a multifaceted quantification method that analyses the urban water supply security (UWSS) on a case study basis (Krueger, Rao, & Borchardt, 2019). UWSS is a rising concern for an increasing amount of metropolitan regions worldwide (He et al., 2021). The CPA framework assists in identifying significant and influential aspects that give a more holistic impression of the current state of a city's UWSS. The CPA will do so by analysing five 'capitals' that focus on the interaction between social, ecological, and technological systems (SETS).

UWSS is affected by the amount of water supply (natural capital; N), the infrastructure in place to transport the water to the individual households (physical capital; P), the monetary means to support the system (financial capital; F) and the political aptitude to strengthen and reinforce the necessary actions (institutional capital; I). Where and if these four aspects lack in the ability to support all inhabitants in the access to safe water, the community will try to bridge the gap. This ability will be quantified as social capital (A). The five capitals are analysed across three dimensions: availability, robustness, and risk. The availability aims to quantify the current presence of the different capitals in the system. The robustness defines the ability to buffer shocks, recover after receiving a shock and the insensitivity to disturbances that result from risks. This leads to stable system performance in spite of variability in system components (Krueger et al., 2019; Mens, Klijn, de Bruijn, & van Beek, 2011). The dimension risk reviews a range of risk factors and categorises whether

it is likely for the risk to occur in the system. The exact indicators used for the dimensions measured in the CPA are shown in the methodology section.

2.1 History of Mexico City

This research will adopt the CPA framework to the Federal District of Mexico City (abbreviated: Mexico City). This is the official name of and demarcation for the political and administrative entity that operates relatively independently from other states and has large autonomy (Reyes, 2019). This region does not include the complete urban agglomeration surrounding Mexico City, also called Metropolitan Area of the Valley of Mexico, or Greater Mexico City. The focus in this research is on the former due to clearer and less complex availability of administrative and scientific data. For contextual understanding of the region and its historical development, this distinction is less apparent. Greater Mexico City is the fifth most populated urban area worldwide, with the current population of 22.3 million inhabitants, of which 9.2 million live in Mexico City (UNDESA, 2018b; INEGI, 2020).

The Aztecs, the first settlers of the current capital city arrived in the 14th century and lived in Tenochtitlán, an island in Lake Texcoco with a population of about 200.000 people (Gutiérrez, 2015). For over two centuries, the Aztecs were able to sustainably manage their water resources by using the surrounding lakes. Later, the Spanish colonisation caused the fall and destruction of Tenochtitlán and made way for the capital of New Spain (nowadays Mexico) to be build on the same place (Gutiérrez, 2015). Throughout the years that followed, the population grew, and the city expanded, see Figure 1.

With the arrival of the Spanish, they quickly discarded the Aztec sustainable water management to create more space for the expansion of the city. Ultimately, the Spanish drained the lake to start building upon the lakebed (Deltares, 2016). From this pivotal moment onwards, Mexico City was subjected more frequently and systematically to water-related problems due to the change in land and water management, challenges that are still felt today (Montero-Rosado et al., 2022). Nowadays, the former lakebed is in the heart of the metropole. To meet the rising demand of water, the Mexican government constructed the Lerma and Cutzamala water systems between 1965-1992 (SACMEX, 2021).

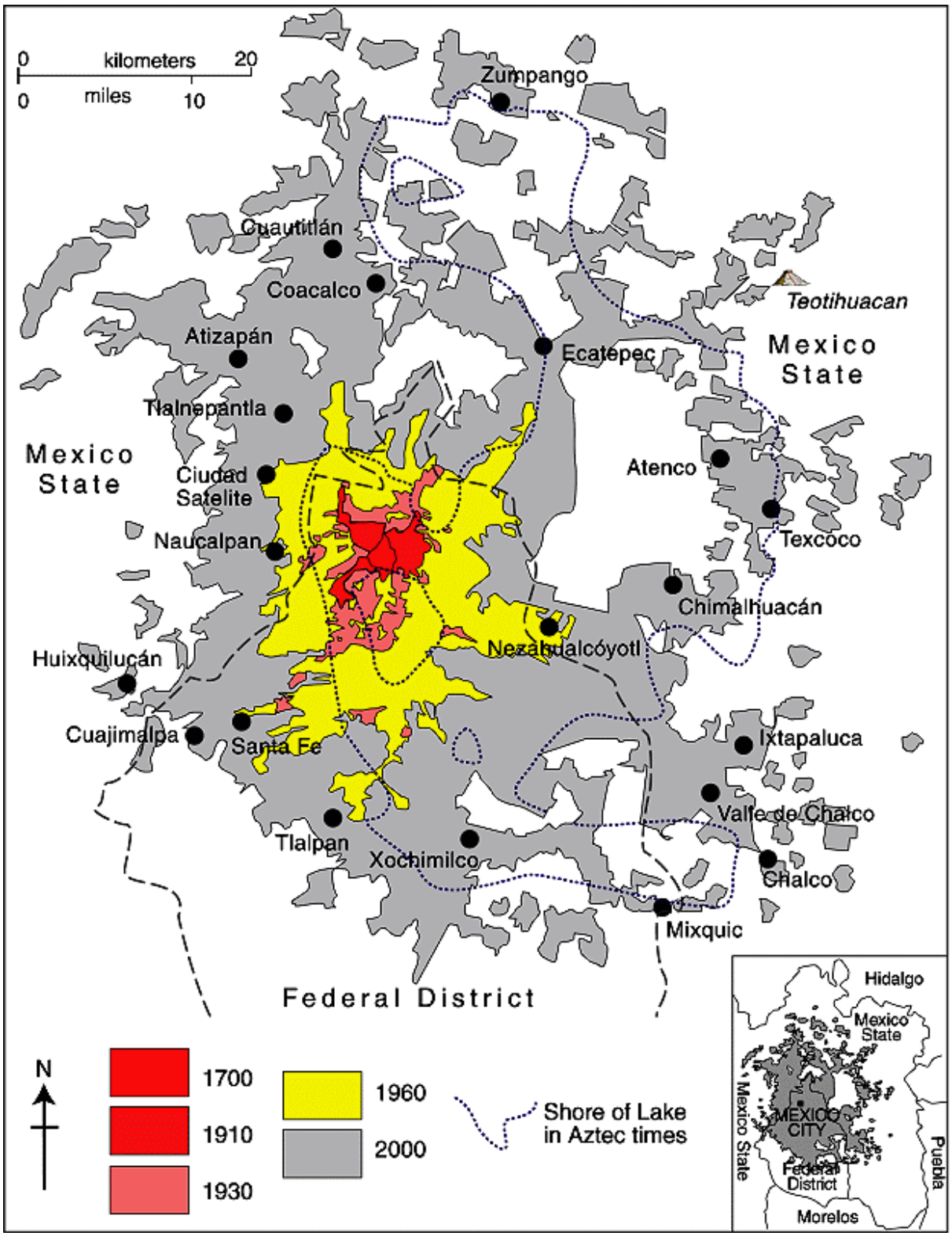


Figure 1: The spatial expansion of (Greater) Mexico City over time, from 1700-2000. Source: Capron, et al. (2017).

2.2 Geophysical factors

The hydrogeography of the region impacts the susceptibility and resilience of the water supply system to floods, low water quality and earthquakes to a great extent. Mexico City is located in a valley that serves as an endorheic basin. This is a closed drainage system in which a natural outflow of water is not possible (Palma, Rivera, & Carmona, 2022). This makes Mexico City susceptible to floods and inundations after precipitation events during the rain season (Deltares, 2016). 66% of the water demand in Mexico City is provided by the extraction of groundwater from the aquifers underneath the city's soil (Palma et al., 2022). Overexploitation of the aquifers has caused the land to subside to rates up to 423 mm/year, effects which are noticed throughout the metropole as seen on the front page of this document (Fernández-Torres et al., 2022). The overexploitation compacts the lacustrine sedimentary soil above the aquifer (Madrigal, Botero, & Díaz-Ávalos, 2022). This causes a weakening in the recharge ability of the soil to absorb water after precipitation events (Han & Cao, 2018). Therefore, the overexploitation of the aquifers decreases both the long-term availability, and the resilience of the city's water resources. Figure 2 shows the water budget in the Mexico Basin and gives an insight into the proportionality of the water attributed to each aspect in the balance. As mentioned above, the pumping rate exceeds the recharge rate. Also, the streamflow out is a result of an anthropogenically constructed drainage system (Montero-Rosado et al., 2022).

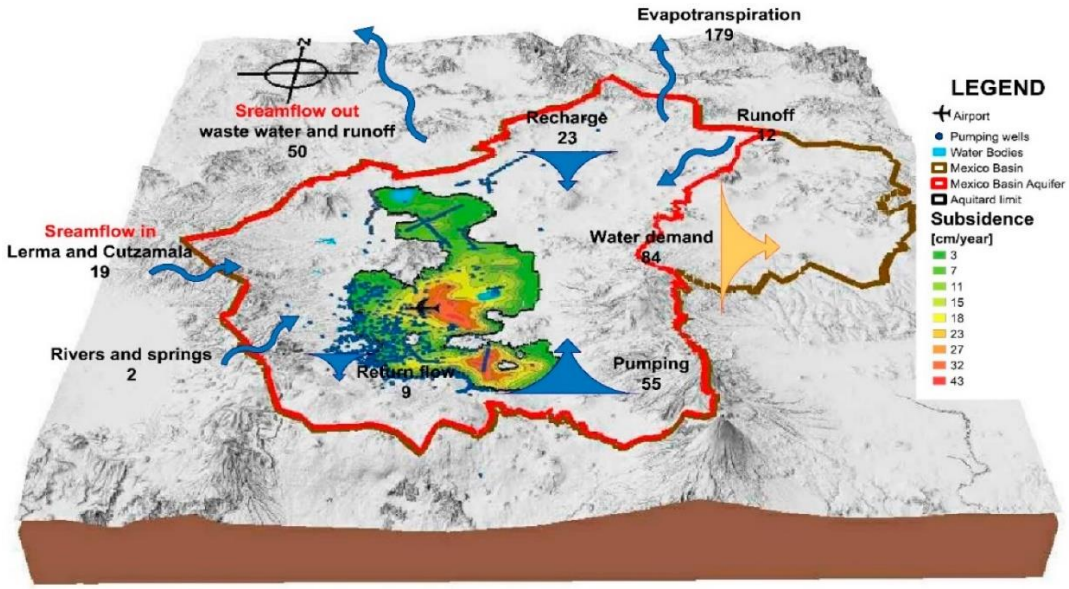


Figure 2: The water budget of the Mexico Basin, comprising the Federal District of Mexico City and the majority of the metropolitan region surrounding it. The land subsidence rates are shown in a green-red continuum. The blue arrows represent the different types of hydrological movement in the system. The orange arrow represents the extraction of water from the system. Source: Palma, et al. (2022).

The geology of the region can be divided into two distinct aspects that impact Mexico City. Firstly, the geological base is distinctly different between the mountains surrounding Mexico City and the majority of the city's surface area. The mountains and hills surrounding the city mainly consist of andesitic and dacitic rock material (Arce et al., 2019). These are both igneous rock types and do not erode or weather easily. The geological base of the urbanised area – the former lakebed – consists mainly of lacustrine deposits, which are sedimentary layers of clays, silts and sands (González-Hernández, Carreón-Freyre, Gutierrez-Calderon, Cerca, & Flores-Garcia, 2015). This is a soft soil type made up of weathered and deposited materials. For this reason, the central areas of Mexico City are more affected by land subsidence than the suburban and rural areas further up the mountains.

Secondly, the Cocos Plate subducts underneath the North American Plate at the Southwest coastline of Mexico, near the state Michoacán. Due to this fault, two earthquakes have hit Mexico City in the recent past, in 1985 (M_W = 8.1) and 2017 (M_W = 7.1) (UNAM, 2023). The area is susceptible to more seismic activity in the future, either from the volcanic plates, or from crustal faults in the Trans-Mexican Volcanic Belt (TMVB) (Novelo-Casanova et al., 2022). The TMVB stretches across the country from 18°30'N and 21°30'N and passes through Mexico City (Ferrari, Orozco-Esquivel, Manea, & Manea, 2012). The 2017 earthquake has shown to negatively impact the water infrastructure and quality in different districts in Mexico City (Espinosa-García et al., 2015; Mazari-Hiriart, Tapia-Palacios, Zarco-Arista, & Espinosa-García, 2019).

The two geological conditions combined pose a cumulative risk to Mexico City. The seismic activity and the soft, water-saturated lacustrine soil can cause liquefaction of the soil (Sana, 2020). This means that during a seismic event the soil transforms into a liquid mass and cannot support the foundations of the buildings on top of it any longer. These are risks that should be taken into account to increase the robustness of the UWSS in Mexico City.

2.3 Socioeconomic factors

Within Mexico City, there are large disparities between the socioeconomic statuses and living conditions of the inhabitants. Impoverished and more affluent neighbourhoods can alternate rapidly throughout the metropole, see Figure 3. The large intracity differences have several causes and outcomes that contribute to instability in the UWSS for the capital's inhabitants. Mexico (official name: United Mexican States) scores 31/100 on the Corruption Perceptions

Index (Transparency International, 2022). Corruption has large effects on the societal and economic development of a country (Dimant & Tosato, 2018). It impacts institutional, jurisdictional, and economic decision-making and can thereby favour individual gain over collective benefits. Mexico is an electoral republican democracy. In practice, democratic values are under high pressure. Civil liberties are systematically restricted, high impunity levels exist amongst government officials and separation of power within the trias politica is skewed, where the elected president enjoys king-like power (Flores-Macías & Zarkin, 2022; Le Clercq & Sánchez Lara, 2016; Villanueva Ulfgard, 2023). These developments have negative implications for the UWSS in Mexico City, for example regarding the efficient distribution and inclusive accessibility of water resources.

58% of all the water supply is retrieved within the administrative borders of the Federal District of Mexico City. The remaining 42% has its origin outside of these borders, from either the aforementioned system Cutzamala (30% of total supply, 130 km west of Mexico City, in Michoacán) or system Lerma (12% of total supply, 40 km southwest of Mexico City, in Estado de México) (Rodríguez-Tapia, Revollo-Fernández, & Morales-Novelo, 2017). Lack of adequate infrastructure, water pressure and political will causes infrequent and unreliable water supply in the – poorer, and more densely populated – Southern and Eastern neighbourhoods of Iztapalapa and Xochimilco, further away from the water sources (Huberts, Palma, Bernal García, Cole, & Roberts, 2023).

Mexico City has an extensive water supply infrastructure in which 98% of the households is connected to the public water supply network (INEGI, 2018). However, the water quality throughout the city, and generally throughout the entire country, does not meet the standards of drinking water (Espinosa-García et al., 2015). Therefore, most inhabitants are forced to provide in their own drinking water consumption, through the purchase of bottled water or by filtering the tap water (Rodríguez-Tapia et al., 2017). The water system in Mexico City is managed by the semi-governmental institution *Sistema de Aguas de la Ciudad de México* (SACMEX). Their financial budget depends on income from water tariffs and government subsidies attributed to SACMEX. However, SACMEX fails to cover all costs due to insufficient government investments and high amounts of non-revenue water from leakage, illegal tapping and administrative shortcomings to collect all water tariffs (Tortajada, 2008). All factors combined contribute to the distrust Mexican society has in the urban water system and governmental conduct in general (Huberts et al., 2023).



Figure 3: Wealth inequality in the Santa Fe neighbourhood in Mexico City. Large socioeconomic differences also contribute to large intracity differences in urban water supply security. Source: Miller (2022).

2.4 Academic relevance

This novel multidisciplinary quantification analysis of the UWSS in Mexico City sets the groundwork for an integrated approach to tackle persisting problems in this metropole. The importance of this research is threefold, and expresses itself on various levels.

Firstly, this research adds to the increasing attention to interdisciplinary research and its merits. Interdisciplinary research, in contrast to disciplinary research, can simulate a complex system of interacting aspects and disciplines that would otherwise not be considered. The analysis of this case study informs us more about how the current state of the water system is, instead of a detailed analysis per discipline. In reality, disciplines are not separated and the CPA framework, with this research as its product, shows that. This research can help all involved actors to understand the strengths and the pitfalls of the system at large. Mexico City has broad and wide-ranging pressures (e.g. urbanisation, climate change and corruption) to which it is subjected and influenced by. This research aims to fill knowledge gaps that come forth from the system dynamics, pressures and interactions of the different

disciplines, also called ‘capitals’. Knowledge gaps exist regarding systemic interrelations of – by the CPA framework – identified leverage points.

Secondly, the critical assessment and the tailor-made changes to the CPA framework can also be used to further improve the framework, by making it more inclusive, detailed and interconnected. This will then lead to a methodology that is better able to simulate and understand the UWSS of an urban area. This second aspect is more focused on improving the application of the methodology, in an attempt to increase the usage of the CPA framework in this field of research.

Thirdly, this research contributes to comparative quantitative research on UWSS. Eventually, this will lead to a body of research on different metropolitan areas worldwide, with possibly vast differences, but with potentially overlapping socioeconomic, geophysical and/or political conditions. In turn, this can strengthen international collaboration to tackle similar UWSS-related problems.

Overall, this research will attempt to answer the following question: *How can a critical assessment of the capital portfolio approach quantify and analyse leverage points in the urban water supply security in the Federal District of Mexico City?* Mexico City is subjected to a myriad of interacting pressures that can lead to detrimental effects on the UWSS in the upcoming decades. This research will point out the impactful actors and thereby illustrate the current condition of the urban water supply security in Mexico City. The goal is to understand the interactions between the five different capitals and how they contribute to a holistic overview of the UWSS in Mexico City. Especially for policy makers, it is important to understand the full framework in which decisions play out, and how changing specific aspects of the water security system influence the system at large. This research tries to contribute to this insight and give a more comprehensive tool in understanding the complexity of UWSS. The hypothesis is that the UWSS will not be covered by the public services and that the adaptive capacity of the inhabitants is needed to cover in their basic water needs. Also, governmental interventions are not likely to follow through on the changes necessary, due to high levels of corruption and impunity. Moreover, expected is that the quality of the infrastructure will be one of the main bottlenecks in the system, due to its connectivity to many of the problems related to the UWSS, such as land subsidence and sustainable management.

3. Methodology

In 2019, Krueger et al. constructed a methodology that proposes a framework to quantify the UWSS in any given urban area. This methodology is called the Capital Portfolio Approach (CPA). By quantifying the UWSS through the CPA, the possibility arises to compare cities that differ widely in socioeconomic, geophysical and/or political backgrounds. The five 'capitals' of the CPA framework are operationalised into 66 different indicators (Krueger et al., 2019). The framework aims to quantify the cities on the aforementioned three dimensions: availability, robustness and risk. All three of these dimensions are derivatives from the operationalised five capitals. The three dimensions include areas of uncertainty. These are included to take into account discrepancies in data values and possible deviations from the educated estimations in the CPA framework. The availability works with both a lower and higher bound. For robustness only the lower bound is included in which the calculated value serves as the outer bounds of system robustness. This includes the possibility for a less robust system. The risk dimension works only with the higher bound of uncertainty to include the possibility for higher susceptibility to risks.

Table 2 shows an extensive overview of all indicators, their definitions and values. Remaining unclarities and extra information on the exact calculations, conversions and detailed explanations that are made to quantify every indicator can be found in the supplementary information in Krueger, et al. (2019).

Table 1: Operationalisation of the capitals in CPA. Source: Krueger, et al. (2019).

Type of capital	*Indicators*	*Calculation method*
Natural (N)	UWA (all sources of water)	$N = \frac{\frac{UWA}{\#\ of\ inhabitants}}{water\ scarcity\ threshold}$
	Urban water scarcity threshold (100 $m^3 cap^{-1} y^{-1}$)	*Ratio of N indicates a category of water scarcity.*
Physical (P)	Fraction of houses connected to public water supply system (h)	$P = h * Sw - q * Wdrink$
	Fraction of UWA delivered to houses (S_W)	$Sw = \frac{UWA - Wleakage}{Dw}$
	Leakage of the infrastructure ($W_{Leakage}$)	$Wdrink = \frac{7.3 * population}{Dw}$
	Water demand (D_w)	
	Drinkability of public water supply (q)	
	Ratio of drinking water demand / total water demand (W_{drink})	
Financial (F)	Expenditures in water sector	

	Income in water sector	$F = \frac{income}{expenditure} * P$
Institutional (I)	Institutional efficiency	*These three indicators include 12 binarily assessed metrics. These will be arithmetically aggregated and result in a number between 0-1.*
	Accountability	
	Regulatory complexity	
Social (A)	Additional water sources accessed by the community (W_{extra})	$A = \frac{Wextra}{Dw} + g * \left(1 - Sw + \frac{Wextra}{Dw}\right) + q * Wdrink$
	Supply gap bridged by residents, through storage, rationing and water treatment at home (g)	

When the term 'low' or 'high' is used in the framework, this refers to alternative scenarios connected to a particular indicator. The reason for this could be due to large intraregional differences and the connected differences in data outcomes. The terms 'low' and 'high' are also used to nuance certain indicators, when its value is not conclusive. These values compose the abovementioned area of uncertainty. In this case study for example, the term 'low' is often used to distinct data related to poorer neighbourhoods in Mexico City, such as Iztapalapa and Xochimilco. The term 'high' is often used related to more affluent neighbourhoods, such as Miguel Hidalgo and Santa Fe. This reflects the range of UWSS within Mexico City, and at times is the result of uncertainty in data. The calculations that follow from these combined information sources give a quantified view on the UWSS in Mexico City. The outcomes of this quantification method will help identify the strengths, weaknesses and leverage points for improvement in the UWSS system.

Table 2: Indicators and their definitions of the indicators.

Indicator	*Definition*	*value*
Water Availability [MCMy]	The average amount of water provided by SACMEX. This water flow from source to consumer is measured at the source, thereby including the leakages that occur later in the transportation chain.	*1009.75 MCMy*
UWA high [MCMy]	The total amount of water that can be administered to the water system. This is determined by triangulation of data concerning average amount of water administrated per capita. This water is provided by both SACMEX and alternative water sources, such as cisterns.	*1075.73 MCMy*
Population [people]	Amount of inhabitants in the Federal District of Mexico City. The focus of this research is on this administrative entity due to the clearer demarcation and availability of data.	*9.21 million*

Demand [MCMy]	Total amount of water needed to meet the demand of all inhabitants in the Federal District of Mexico City.	*595.01 MCMy*
Storage capacity [MCMy]	The amount of water that can be used from storage in a sustainable way. This is defined as the recharge rate of the aquifers and includes the current storage capacity of the Cutzamala system. The Mexico City and Lerma aquifers serve as large subterranean reservoirs and can be called upon when needed. However, the actual pumping rates and/or total aquifer content are not used, as this does not fit into a sustainable long-term UWSS strategy.	*1073.2 MCMy*
Water supply [MCMy]	This is the amount of water after leakage and administrative losses to the system have been deducted. This is the amount that is actually administered to the households in Mexico City.	*584.015 MCMy*
Transboundary sources [MCMy]	Water sources that are located outside of administrative and/or political jurisdiction. In this case, this does not exist, as the sources which are geographically outside of Mexico City, are administratively linked to it.	*0 MCMy*
In-boundary water supply [MCMy]	All water supply sources which are inside of administrative and/or political jurisdiction. In this case, this includes the Mexico City aquifer, surface waters and the Cutzamala and Lerma systems.	*1009.75 MCMy*
Source diversity	The amount of different water sources and types of water sources the urban region uses to meet the water demand. Measured on a scale 1-4.	*4*
Water quality control	The extent of the water being tested and up to qualitative standards before public consumption, measured on a scale 1-4.	*2*
Wastewater treated [frac.]	The fraction of wastewater which is treated after public consumption. The threshold value for water to be considered 'treated', is secondary filtration. In this stage, suspended solids, biodegradable organic matter and chemical oxygen demand are filtered out of the water.	*0.1975*
Pop. covered by sanitary infrastructure [frac.]	The fraction of the total population that is connected to the public sanitary infrastructure.	*0.98*
Households connected to water infrastructure [frac.]	The fraction of the total amount of households that is connected to the water supply infrastructure.	*0.98*
Water leakage [frac.]	The fraction of the total amount of available water which is unaccounted for due to water leakage.	*0.42*
Leakage upper [frac.]	The minimal fraction of leakage in the water system. This number represents a better-case-scenario due to the continuous improvements	*0.38*

	of the hydraulic infrastructure.	
Delivers drinking water [0/1]	Binary representation of the water quality. If the delivered tap water is potable, value is 0. If not, value is 1.	*1*
Anticipatory maintenance [0/1]	Efforts to conduct preventive maintenance to the water system to avoid future reactive maintenance. If so, value is 1.	*0*
Emergency solutions for power failures [0/1]	Plans available to implement emergency measures to uphold the water system in times of power failures. If so, value is 1.	*1*
Continuous water supply [0/1]	The uninterrupted provision of water to all neighbourhoods of the urban area. If so, value is 1.	*0*
Monitoring for leakage detection [0/1]	The presence of continuous monitoring in the hydraulic infrastructure to proactively detect leakages. If so, value is 1.	*0*
Average materials age < 50 years [0/1]	The average age of the hydraulic infrastructure since instalment. If younger than 50 years, value is 1.	*0*
Redundancy of critical nodes [0/1]	The presence of the amount of critical nodes in the system. More critical nodes will increase the flexibility and adaptability of the system, when supply in the hydraulic system is intermittent or unreliable.	*1*
Possibility of emergency zone isolation [0/1]	The ability of the hydraulic urban system to close off areas individually. Thereby the spread of potential contaminants and/or disturbance during infrastructural maintenance is limited. If so, value is 1.	*1*
Modularity source-to-sink [0/1]	The flexibility and adaptability of the water system to adjust to changing supply and demand, thereby serving the inhabitants in need of water supply. If solid plans to ensure this are implemented, value is 1.	*0*
Income [millions of MXN]	The total amount of money, in Mexican pesos, budgeted for SACMEX, through both tariffs and subsidies. This is the averaged annual income between 2016-2022.	*13975.86*
Spending [millions of MXN]	The total amount of money, in Mexican pesos, actually spent by SACMEX, on construction and maintenance of infrastructure and wages. This is the averaged annual spending between 2016-2021.	*14720.48*
Cost recovery [0/1]	Cost recovery reflects the (in)dependence of the water system to government subsidies and/or donor funding. The higher the fraction of income can be traced back to water tariffs and fees, the less dependent the system is on external funding, thus has the ability to operate more independently.	*1*

Medium-income city [0/1]	The mean income of inhabitants of Mexico City, set against the World Bank middle income threshold value of 1026USD/year. If met, value is 1.	*1*
Energy autonomy is >50% [0/1]	Energy autonomy is met when more than 50% of the energy can be obtained within the administrative/ political borders, the Mexican borders in this case. This decreases the dependency on other political entities. If so, value is 1.	*1*
Clear communication protocols for information sharing [0/1]	The presence of communication networks by SACMEX that are able to spread information clearly to the desired recipients. If so, value is 1.	*1*
Institutional feedback loops [0/1]	The administrative ability of SACMEX to receive feedback from involved stakeholders on existing flaws and through that adopt new measures to improve. If so, value is 1.	*0*
Customer complaints follow-up mechanisms [0/1]	The ability of SACMEX to follow-up on customer complaints, by having a customer service available, staffed and able to act upon the received complaints. If so, value is 1.	*0*
Corruption Perception Index > 50 [0/1]	The country index provided by www.transparency.org, which measures the corruption perception index, out of 100. The higher the value, the more trust is perceived in the country. If the value is higher than 50 , the value is 1.	*0*
Administrative losses < 10% [0/1]	The amount of unaccounted-for water exiting the water system, for which SACMEX does not receive revenue. This excludes the amount lost by leakage. If percentage is < 10%, value is 1.	*0*
Affordability < 5% of income [0/1]	The median percentage of income a household spends on water. If this is less than the threshold of 5%, the value is 1.	*0*
Urban-urban strategies [0/1]	The implementation of strategies to combat water-related challenges that concern the urban area (different municipalities) within the metropolitan area, hereby making use of collaboration between institutions and/or companies. If so, value is 1.	*1*
Urban-rural strategies [0/1]	The implementation of strategies to combat water-related challenges that connect the urban area with the surrounding rural areas. This happens through collaborative and/or communicative procedures. If so, value is 1.	*1*
Mechanisms for groundwater management [0/1]	The implementation of mechanisms and the ability to execute plans to manage groundwater resources. If so, value is 1.	*1*

Mechanisms for surface water management [0/1]	The implementation of mechanisms and the ability to execute plans to manage surface water resources. If so, value is 1.	*1*
Emergency operations planning [0/1]	The existence of emergency operations planning regarding water availability and security in the case of emergency situations. If so, value is 1.	*1*
Inter-sector coordination in operation and management [0/1]	Existing collaboration between other influential sectors, such as energy, wastewater and logistics. Coordination between these involved stakeholders improves efficiency and overall performance of the water system. If so, value is 1.	*0*
Capacity to improvise, innovate and expand operations [0/1]	The administrative and institutional ability to respond to circumstances in which swift adaptation to emerging situations is necessary. A measure of institutional flexibility and quick adjustability. If so, value is 1. A more elaborate operationalisation of this indicator can be found on page 23.	*1*
Implements training & innovation for resilience and sustainability [0/1]	The existence of training, educational programmes to improve community resilience and/or assistance in (individual) interventions to improve the hydraulic system's functioning. If so, value is 1.	*1*
National support planning [0/1]	The existence of a framework or plan to assist and guide in recovery from disasters. If so, value is 1.	*1*
High city ranking [0/1]	The importance of the urban area, with regard to emergency response and assistance and disaster relief from the government or foreign actors. If so, value is 1.	*1*
W_{extra} [MCMy]	The annual amount of available water accessed by the inhabitants outside of the public water supply system.	*76.40*
Sw [MCMy]	In the case of Mexico City, Sw is equated to the public water supply.	*584*
Average supply days	The averaged amount of days per week water is supplied to households, through the public water system.	*5.922*
Median income > " middle income" [0/1]	If the median income is higher than the middle (mean) income, it gives an indication of the distribution of wealth. If the median is higher, the distribution is more even and the value is 1.	*0*
Access to alternative water ways [0/1]	The ability for households and/or individuals to obtain alternative sources of water, outside of the common water supply system. If so, value is 1.	*1*
Has storage capacity [frac.]	The ratio of households that have the ability to retain a quantity of water to self-sustain for at least one week (7 days).	*0.22*

Information availability on system functioning [0/1]	The presence of clear communication regarding the up-to-date status of the hydraulic system and infrastructure, such as when infrastructure maintenance or water deliveries take place. This can happen through traditional media, social media and/or other methods of communication. If so, value is 1.	*1*
Active community structures [0/1]	The ability to cooperate, assist and support other members in the surrounding community. This improves the overall social cohesion and decreases individual vulnerability to water insecurity. If so, value is 1.	*1*
Refrains from drinking contaminated tap water [frac.]	A fraction of the amount of inhabitants that uses alternative water sources when the public services are not of sufficient quality.	*0.43*
Direct access to water sources [0/1]	The ability for households to gather, obtain and/or harvest water without the intervention of the public water supply system. If so, value is 1.	*1*
Acute geophysical hazards (e.g. earthquakes) [0/1]	The occurrence of acute geophysical hazards, such as earthquakes, landslides, tsunamis and/or volcanic activity. If so, value is 1.	*1*
Chronic geophysical hazards (e.g. land subsidence) [0/1]	The occurrence of chronic geophysical hazards, such as land subsidence and/or sea level rise. If so, value is 1.	*1*
Socio-economic and political changes [0/1]	The occurrence of socio-economic and/or political changes that could influence the stability and long-term vision of the urban area and/or country. If so, value is 1.	*1*
Immediate threat of war/terrorism [0/1]	The occurrence of war/terrorism occurring in a short timeframe in the urban area. If so, value is 1.	*0*
Illegal tapping into water pipes [0/1]	The occurrence of illegal tapping and/or other criminal activity to obtain water from the hydraulic system illegitimately. If so, value is 1.	*1*
Economic crises [0/1]	The occurrence of economic crises in the urban area and/or country. If so, value is 1.	*0*
Competition for resources [0/1]	Competition of water resources with other industries or urban/rural areas, thereby making the water availability to the case study area less certain. If so, value is 1.	*1*
Risk from industrial spills [0/1]	The occurrence of industrial spills that can contaminate the hydraulic system, thereby endangering safe water supply and administering to the population. If so, value is 1.	*0*
Risk of epidemic incidents due to	The occurrence of degraded and outdated infrastructure, which brings about higher rates of epidemic illnesses in the water, such as hepatitis and typhus. If so, value is 1.	*1*

infrastructure degradation [0/1]		
Storms or wildfires [0/1]	The occurrence of storms and/or wildfires that damage the hydraulic infrastructure. If so, value is 1.	*0*
Floods or droughts [0/1]	The occurrence of floods and/or drought that limit the water availability and/or effluent capacity in the urban water management. If so, value is 1.	*1*
Extreme temperatures [0/1]	The occurrence of high temperatures in the climate of the urban area. This can lead to faster degradation of hydraulic infrastructure, due to e.g. freezing of pipes. If so, value is 1.	*0*

3.1 Getting to the results

The Supplementary Information template (see Appendix) is used for the quantification phase of the case study. The Excel sheet divides the different capitals and dimensions coherently, and highlight the required cells to be filled out. This template is both important and convenient to use, as it ensures a standardized methodology and quantification of the indicators. Over time and with the analyses of numerous urban areas, this will lead to a strengthened and more reliable methodology to quantify UWSS.

Filling out the indicators can be a tedious task. Indicators are operationalised thoroughly and have exact definitions, which should be adhered. More often than not, certain terminologies such as water supply, demand and availability are used interchangeably in scientific and/or policy reports. It is important to draw attention to the exact definitions, to guarantee a sound analysis of the right indicators. Moreover, depending on the location of the urban area, data availability and data reliability can be a limiting hurdle. Tailor-made adaptations to the framework can at times be the only way forward. In this case study I was compelled to make several changes, which will be mentioned and explained in section 3.2.

To ensure a high standard of information quality, it is advisable to make use of multiple sources that can verify the quantification of the indicators. This limits the margin of error and the range of uncertainty. If availability allows, use a diverse palette of sources to limit information bias. The results from the filled-in CPA framework template give us a rough overview of the dimensions and the corresponding capital values. The next step is to identify the strengths and the leverage points of the system. Leverage points are points within this

system that can have large impact on the functioning of the system, either good or bad. At the same time, these leverage points can be considered the weaknesses if already implemented, but not enforced throughout time. Identifying the leverage points happens through trial-and-error. When all values are determined, it is useful to change values one-by-one and see what the impact of these changes are to the capital values. The most influential changes are the leverage points of the system.

3.2 Interpretations of and changes to the indicators

The CPA framework is, as aforementioned, the general template to calibrate all cities to the same standards of UWSS. However, due to a great variety in data availability and city-dependent environmental factors, some inconsistencies, preventable uncertainties or blanks might arise in the framework. Here, I will explain and justify the deviations I made from the CPA framework, with the aim to optimise its functioning.

Storage capacity is defined as the total amount of water (in MCMy) that is stored within the administrative area and can be used to assist in meeting the water demand when necessary. Mexico City is built on top of an aquifer and extracts water from aquifers in the Lerma system (Rodríguez-Tapia et al., 2017). Therefore, all the water in these aquifers could potentially be used to aid in times of water scarcity, as it is in their right and at their disposal. However, this unsustainable method of water management would lead to further land subsidence, damaged hydraulic infrastructure and reduced the resilience of the water balance in the urban area (Chaussard, Havazli, Fattahi, Cabral-Cano, & Solano-Rojas, 2021). Less water can infiltrate back into the aquifers the more water is pumped out, due to the compacting soil of the lacustrine bedding (Palma Nava, Parker, & Carmona Paredes, 2022). Inflating the storage capacity to this amount would give the illusion of an unrealistic and non-existing buffer. Therefore, I use the recharge rate of the aquifers as the storage capacity. This rate is linked to the sustainable use of the aquifer, as the recharge rate is negatively impacted by the overextraction rate. Also, this is the amount that can be extracted yearly without worsening the environmental effects of overextraction.

Trans- and in-boundary water availability covers the dependency on external administrative and/or political entities. As mentioned in section 2.3, water supply in Mexico City is retrieved from in-boundary aquifer sources, and the transboundary Cutzamala and

Lerma system. However, this hydraulic system is constructed to meet the demand in Mexico City and all the water is automatically transported towards the urban area (*Sistema De Aguas De La Ciudad De México, n.d.*). There is competition for these water sources between several indigenous groups in the local area and the large-scale transport to Mexico City (Campos-Cabral & Ávila-García, 2015). These local indigenous groups have progressed in obtaining their communal water rights, but there is no broad administrative or political dispute about these resources (*Vatterott, 2018)*. The reliability in obtaining this water and supporting the demand in Mexico City remains high. Therefore, these sources are considered in-boundary sources, adding it all up to 100% in-boundary source.

The spent money by SACMEX is published annually in October, for the previous year, by the Administration and Finance Secretariat of the Government of Mexico City. Therefore, the latest data on spent amount by SACMEX in 2022 is not yet available and the budgeted amount is used instead.

Cost recovery gives an indication of the financial independence of the hydraulic system. The system is partly dependent on income originating from water tariffs and bills, and partly on government subsidising. Even though there is a dependency on the government, there is also a mutual benefit to a well-functioning hydraulic system. At the same time, the Mexican government also has influence on the height of the water tariffs put out by SACMEX. Therefore, there is an intricate link between the government and the governmental water institute SACMEX. This relationship is not based on donor-funding or philanthropy, therefore this framework will work with the value '1'.

Administrative losses < 10% is the indicator that includes water theft and other unaccounted-for-water, excluding leakages. This a persistent problem throughout Mexico City's, as water is tapped from public water services and eventually sold on the private market illegally (*VICE TV, 2022*). However, no scientific quantifications have been made that categorise the different kinds of water losses in the Mexico City hydraulic infrastructure. In total, 42% of the administered water is lost between the source and the consumer. The fact that there is no known subdivision between these types of losses, gives the indication that appropriate managerial efficacy is lacking. Without the available data to better understand the loss of 42% of the water in the hydraulic system, the ability to exercise control over the administrative losses decreases. Therefore, this model works with the value '0' for this indicator. Widespread reports of water theft and high non-revenue water rates and the loss

of significant amounts of water point to managerial inefficiency in addressing the accountability (*Luna, 2023*).

The capacity to improvise, innovate and expand operations deals with how both acute and chronic challenges that occur in the water system are handled and combatted. The way this indicator is defined in this framework is: the ability to adapt to the changing needs to maintain or attain a functioning water system. The interpretation in this sense draws back to two sources of evidence. The first one as an indicator for acute flexibility, where the government was able to swiftly act to the 1985 earthquake in Mexico City, making enough resources available to repair the infrastructural damage in a 40-day operation, 24 hours per day (*Sistema De Aguas De La Ciudad De México, n.d.*). The second source regards chronic flexibility, where the government proves to expand operations and address the availability problems in the least connected regions, increasing the service hours of water by 210% in four years time (*Programa De Sectorización*, 2022).

W_{extra} covers the additionally available water sources to inhabitants, when public services are not adequate. Especially in the poorer neighbourhoods, the areas where adaptation to the lack of public services in water supply, it is hard to quantify the amount of water inhabitants retrieve from alternative sources. As stated in Huberts, et al. (2023), one of the main reasons is that inhabitants do not know the amount of litres of water storage they have. However, they are able to indicate the amount of time they can live off the water buffer they have (Huberts et al., 2023). To inductively reason the amount of extra water services inhabitants can obtain, an estimation has been made in the daily usage of water per capita in Mexico City. Together, this number and the data from Huberts, et al. (2023) make up the term for W_{extra}.

The researched group from this article is not an average sample for Mexico City at large. However, it is a group that is disproportionately negatively affected by the limitations of the public water supply system and are above-average able and required to adapt to meet their water demand (Huberts et al., 2023). The public services, reflected by the term CP in the framework, are more adequately managed for other inhabitants in the city, who are less in need of adaptation strategies. The inhabitants receive this water in some cases from governmental *pipas* (water trucks). However, the administering is not reliable and often the water demand has to be compensated and buffered by private pipas, rainwater harvest and recycling of water (Huberts et al., 2023).

Storage capacity over 7 days is used as the household threshold for having enough water supply to withstand shocks and delays in the water supply reliability. This continuity of water supply and the correlating storage capacity differs widely throughout the Federal District of Mexico City. Higher income and more centralised neighbourhoods are less reliant on storage capacity, in comparison to lower income neighbourhoods further away from the city centre, with the latter group having higher adaptation abilities (Huberts et al., 2023). To give more nuance to these stark differences, the value of storage capacity > 7 days is changed from binary (0/1) to continuous. This value is now determined as the ratio of households that have a storage capacity > 7 days (Huberts et al., 2023).

Treats tap water at home indicator is changed to *Refrain from drinking untreated tap water.* Large amounts of households do not drink the tap water, due to the low quality and high risk of contamination. However, many of these households also do not have the means to implement measures to treat the tap water, so are forced to resort to bottled water (Rodríguez-Tapia et al., 2017). This new definition includes both groups of households, those who treat the tap water and those who buy bottled water instead. Also, the value is changed from binary to continuous, to more accurately depict the fraction in Mexican society (WHO/UNICEF, 2020).

Land subsidence is impacting the risk dimension in a number of ways, thereby impacting the water security (Madrigal et al., 2022)**.** As mentioned in the introduction, the recharge rate of the aquifer is reduced due to the soil compaction. This leads to less available natural capital to be pumped up from the aquifer and through that, impacting the capital 'N'. Therefore, this capital has been added as susceptible capital, next to 'P'.

Socio-economic, political changes and unforeseen population growth is an indicator which can be inflated to a broad definition regarding the socioeconomic and political stability of a country. Therefore, this indicator can be highly interpretable. Here, the definition is the stability of the political system and the socioeconomic satisfaction of the population. The indicator of 'immediate threat of economic crises' already covers the economic stability of the country, which is the reason to divert focus from that aspect in this indicator. Mexico is battling various challenges regarding political and socioeconomic stability. There is low trust in government, income inequality is higher than the UN warning level and very high levels of impunity ("Informe País 2020," 2022*; Gini Index*, 2020; *Impunity Levels in the World. Global Impunity Index 2020 (GII-2020)*, 2020). These all have impact on the political and social

instability, which makes the value of this indicator '1'.

A small change to the addition of public services with adaptation (CP+A) has been made, with comparison to the original framework. The current calculation of the higher bound CP+A uses the A_{high}, instead of A_{low}. Adding the maximum adaptive capacity (A_{high}) to the public services (CP+A) does not exceed 1.00, but does give the highest coverage of UWSS possible in this framework.

After the first leg of information gathering and adjustment of the indicators, the ensuing results are calculated in the framework. It is time to analyse the raw data, figure out the meaning of the aggregated numbers and dissect the components of the dimensions and their importance.

3.3 Data Management Plan

Types of data

This research will mostly be based on an interpretation and analysis of previously attained data by other researchers. We will use earlier research that includes observational data and model simulations to base calculations and projections of water supply on. Also, I will base our research on qualitative research relating to community adaptability when water resources are insufficient. Eventually, this research will become an educated mix of several types of existing data that will be used to critically assess all indicators provided by the Capital Portfolio Approach. The mathematical calculations to determine the UWSS is provided in the CPA and will be used to convert the different types of data into one quantifiable scheme.

Contextual details

Firstly, the data necessary to fill the knowledge gaps in the GMC UWSS will be filtered and sorted on a secure drive. These different sources of data will then be sorted systematically in APA-style:

[AUTHOR(S)] ([YEAR]) [TITLE OF THE PAPER], [*NAME OF THE PUBLISHER / WEBSITE*] [SPECIFICS OF THE SOURCE], [DOI / REFERENCE TO THE SOURCE].

From this volume of information, the applicable fragments will be inserted into the programmed .xlsx file provided by (E.H. Krueger, 2022). This file comprises all indicators of the CPA and assists in the mathematical quantification method.

Quality control

An important aspect in conducting high-quality research is to have clear definitions of the mentioned factors and terminology. To determine values for indicators it should be clear what the knowledge gaps exactly are so these can be answered as precisely as possible. Also, the possibility to verify data from different sources is a fruitful way to strengthen the research and raise its quality. There is a possibility that lack of certain data can influence the quality of the research. By administering a margin of uncertainty to this incomplete or partial data, it prevents the data from becoming erroneous. Another verification method is to ask the supervisor to review the data analysis and filter out novice and/or preventable mistakes that impact the quality of the research.

Storage, backup and security

The data will be saved on a secured online server. This prevents the data from getting lost when technical problems occur on one device. The data will be available in as a referral from the final product. From there, it becomes accessible and transparent to the public where the data was produced and aggregated.

Copyright and reuse

As this research will predominantly be based on an application and interpretation of previous research, it is important to dedicate the sources of data accordingly to their rightful owners. The data has been made available to the public and is open-source. Therefore, it is permitted and legitimate to use the data for follow-up research, under the condition that the referencing and citation will be done accordingly. The data that will subsequently be created from this thesis project will be open to examination and will be provided if continuation of research in CPA is desired. During the project, the aggregated data will be viewed by the producer and the two supervisors. After publishing the research, the data will be available to all interested parties and will be provided as supplementary information.

Protection and privacy

There will not be any privacy issues, due to the fact that this research will not conduct any quantitative analyses with individuals, nor will there be new information gathered. Therefore, any protection and/or privacy measures will already have been taken by researchers from which the data used in this research is derivative.

4. Results

The modifications and the data collected have been implemented and inserted into appendix **1** *(the SI Excel sheet Mexico City)*. The next step is to analyse the outcomes of the three dimensions: availability, robustness and risk. These can all be subdivided into the five separate but interconnecting capitals. From here, the strengths and weaknesses of the system are identified. The weaknesses are simultaneously the leverage points, because these are the points that are influential to the system's (dys)functioning. Identifying these points will lead to more knowledge about the impact of weaknesses and how they affect the quantified values of the UWSS.

The radar graphs below visualise the capacities of the capitals per dimension. Every pentagon layer corresponds to 0.25 of capital value. This means that a capital on the outer pentagon has the value 1.00.

4.1 Availability

Table 3: The calculated capital availability values, corresponding to Figure 4.

Capital	Low bound	Value	High bound
N	*1.10*	*1.10*	*1.17*
P	*0.41*	*0.45*	*0.49*
F	*0.28*	*0.43*	*0.66*
I	*0.30*	*0.50*	*0.50*
A	*0.33*	*0.33*	*0.37*
Average CP	**0.39**	**0.54**	**0.62**
Average CP+A	**0.76**	**0.87**	**0.99**

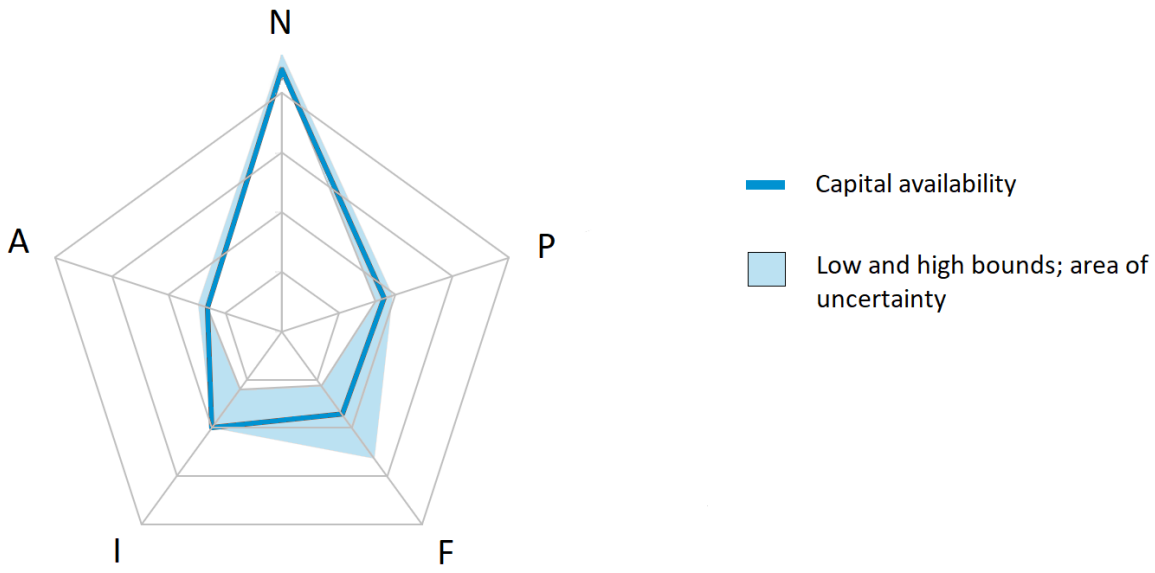


Figure 4: The capital availability values of the N (natural), P (physical), F (financial), I (institutional) and A (social) capitals. The blue line represents the calculated availability values, with the value of the outer line of the radar graph being 1.00, and the inward lines representing steps of 0.25. The light blue area represents the area of uncertainty.

From the lower to higher bounds of uncertainty N > 1.00, meaning that Mexico City has enough water availability per inhabitant to provide the population. The low physical capital (P = 0.45) indicates insufficient capital to support the infrastructural system, largely affected by the high leakage rates. This is closely linked to the financial capital (F = 0.43), where the lack in P necessitates higher investments to cover for the financial deficit. The institutional capital (I = 0.50) indicates that administrative power is not adequate to effectively manage the communicative and collaborative groundwork needed. These four capitals result in the coverage capacity for the public services (CP = 0.54). With CP < 1, meaning that the public services are not covering the necessities of the UWSS and soci(et)al adaptation (A = 0.33) assists. The high social capital means that Mexico City society is able to adapting to the current system and improve their own conditions. However, with the combined value (CP+A = 0.87), it indicates that not yet all necessary services for UWSS are met on the household level.

4.2 Robustness

Table 4: The calculated capital robustness values, corresponding to Figure 5.

Capital	Low bound	Value
N	*0.71*	*0.84*
P	*0.13*	*0.38*
F	*0.67*	*1.00*
I	*0.33*	*0.83*
A	*0.14*	*0.66*
Average RP	**0.46**	**0.76**
Average RP+A	**0.50**	**0.93**

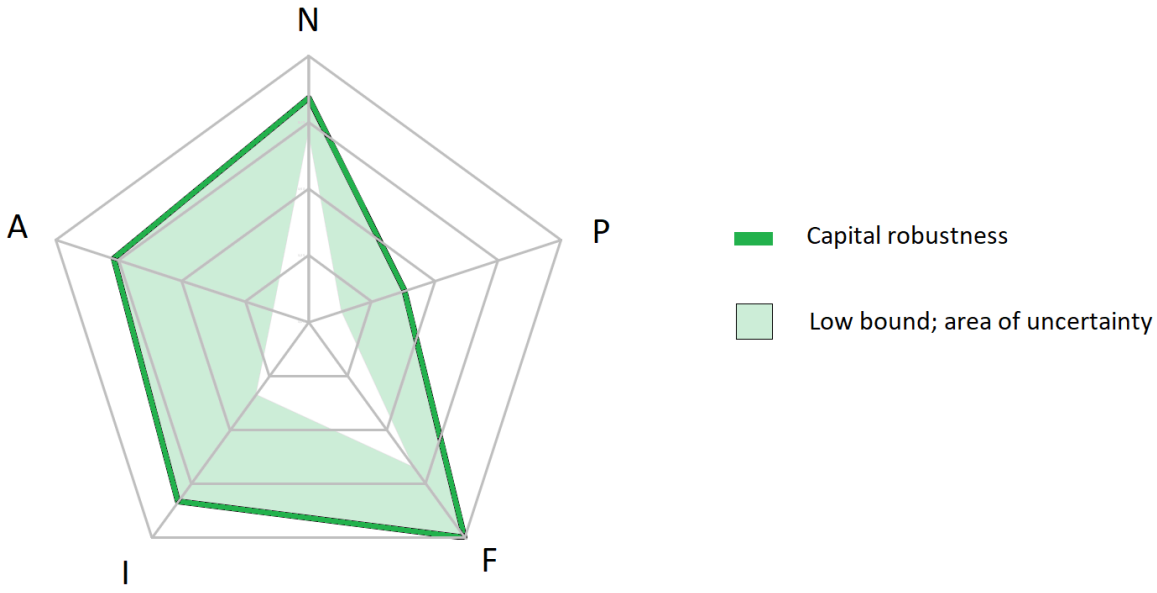


Figure 5: The capital robustness values of the N (natural), P (physical), F (financial), I (institutional) and A (social) capitals. The green line represents the calculated robustness values, with the value of the outer line of the radar graph being 1.00, and the inward lines representing steps of 0.25. The light green area represents the area of uncertainty, on the lower end.

The robustness of the natural resources (N = 0.84) indicates that the system is able to buffer shocks, but still below a desirable level, due to the value being < 1.00. The physical capital (P = 0.38) is a more pivotal element in the system robustness. There is limited ability to adapt to and to buffer changes in the aged infrastructure. Financially, the UWSS has extensive robustness (F = 1.00), covering possible changes in fiscal dependency and energy production.

The institutional capital (I = 0.83) fully covers external robustness and to a far extent also internal robustness. The latter is < 1, where intersectoral collaborative communication is lacking. Adaptive robustness (A = 0.66) reflects the ability of the inhabitants to maintain their interventions to maintain their water security. This is highly dependent on the intraurban socioeconomic differences.

The overall system robustness (RP = 0.76) indicates relative anticipatory readiness, but with specific consideration to the lack of physical robustness. The low bound values indicate the area of uncertainty for decrease in the robustness. This is due to insecurity in coverage of the system if risks increase and higher pressure of the system robustness in poorer parts of the city. The lower bound robustness value (RP = 0.46) indicate a significant reduction (0.30) compared to the overall robustness value. The decrease of robustness also maintains when the social capital is included (low bound RP+A = 0.50; RP+A = 0.93).

4.3 Risk

Table 5: The calculated capital risk values, corresponding to Figure 6.

Capital	**Value**	**High bound**
N	*0.71*	*1.00*
P	*0.56*	*1.00*
F	*0.33*	*1.00*
I	*0.50*	*1.00*
A	*0.38*	*1.00*
Average	**0.50**	**1.00**

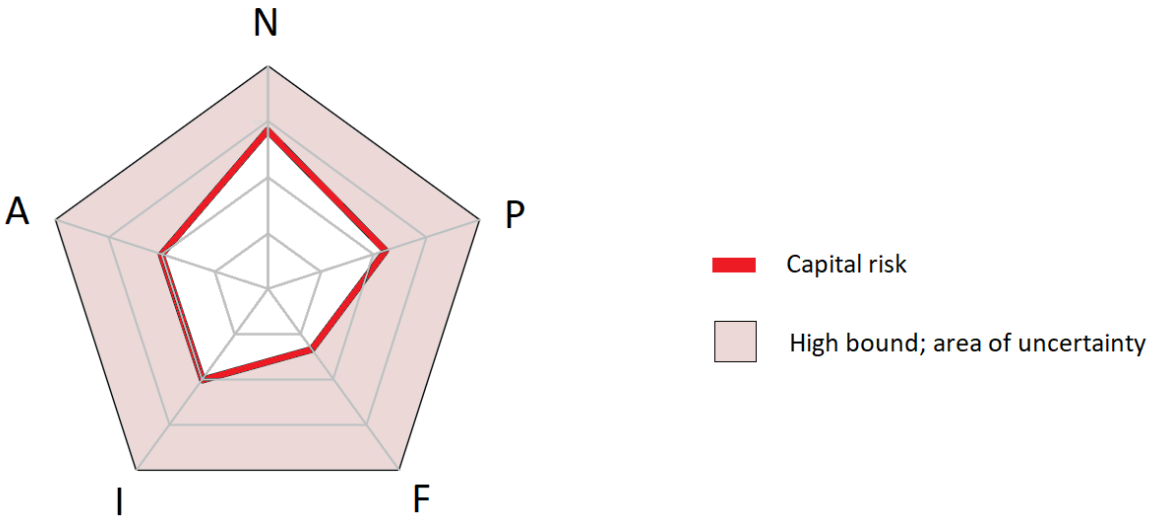


Figure 6: The capital risk values of the N (natural), P (physical), F (financial), I (institutional) and A (social) capitals. The red line represents the calculated risk values, with the value of the outer line of the radar graph being 1.00, and the inward lines representing steps of 0.25. The pink area represents the area of uncertainty, on the higher end.

There is a varying degree of risk throughout the different capitals. Risks reflecting the natural capital (N = 0.71) are considerable, with both chronic and acute risks. The physical capital (P = 0.56), institutional capital (I = 0.50) and adaptation capital (A = 0.38) have a medium level of risk. This means that for capital I, half of the risk indicators from the CPA framework is marked as being possible to occur in the research area. The three capitals also have both chronic and acute risk indicators to which the city is susceptible. The financial risk has the lowest rate (F = 0.33), with one chronic risk indicator that could potentially take place. The higher bound scenario for the risk dimension has more serious implications. Here, all capitals have a risk rate of 1.00, meaning being susceptible to all listed indicators from the CPA framework. With the area of uncertainty reaching 1.00 for all indicators, this means that the calculated values are the minimal risk values.

4.4 Leverage points and weaknesses

Not only were the different capitals and their dimensions calculated, but also the most influential and/or fragile indicators of the system were identified. These are threefold. The first two belong to the dimension risk and impact all five capitals (N, P, F, I and A). These are s*ocio-economic, political changes and unforeseen population growth* and *immediate threat of war/terrorism*. Each indicator has an impact to the risk indicator of 0.19, with the cumulative value being 0.38.

The third indicator with high influence to the CPA framework is the leakage ratio. The amount of water that is lost from source to sink has a large impact on the physical capital and consequently to the financial capital. Lowering the leakage ratio to the hypothetical value 0.10, would mean an increase of physical capital of 0.32 and an increase of financial capital of 0.30.

5. Discussion

5.1 Limitations of the chosen method

This research gives an elaborate and multidisciplinary overview of the UWSS in Mexico City, but is confined within certain boundaries. These boundaries, mentioned here, have the potential to inhibit the reliability, lower the inclusiveness of the research outcomes and simplify these due to complex interacting indicators.

The reliability of the retrieved data is important to discuss critically. Transparency and objectivity can have a different meaning in Mexico – with the country being the most corrupt developed country, according to the World Economic Forum (Kaiser, 2016). This is especially with regard to governmental institutions, and their relation to privatised companies, such as the water distribution companies (Kaiser, 2016). This means that information regarding the financial state of affairs and the conditions of institutional systems in place, among others, can fall victim to the institutionalised corruption. Nevertheless, the information published by government, semi-government or privatised institutions is at times also the only or the most updated source of information available.

The second note to the research results regards to the vastness of and diversity in Mexico City. The megacity includes neighbourhoods with clean running water, where services are fully met and water security is covered on household level. It also includes poverty dominated slums where public services are poor and the role of household adaptation is large. Many other neighbourhoods are along this wide spectrum of urban water supply security. Due to the fact that this research focuses on Mexico City at large, the results may have the effect to split these individual differences. Many scientific articles focus on the specific districts within the city, including the certain extremes, for example in continuity of water supply, which are present in these areas. By extrapolating that research on Mexico City at large, this negates the intracity differences, of for example more affluent neighbourhoods where water runs continuously. Attempts have been made to minimise this by using data from different neighbourhoods, to reflect Mexico City's diversity.

The third limitation is inherently connected to the methodology. The CPA framework uses a multitude of indicators. What the framework yet lacks, are the interconnections between the indicators. Implementing connectivity and causality between the indicators has the ability to resemble system dynamics. For example, corruption is taken into account as one indicator in the availability of the institutional capital. However, corruption (or the lack

thereof) can have widespread implications for financial budgeting, factual representation in government reports and pursual of inclusive and equitable policies. These all influences the effectiveness of necessary system interventions. The CPA framework is a simplified representation of the UWSS, and is not yet able to include the system complexity into its quantification methodology.

Moreover, the research conducted here is transverse, meaning that the information on this case study is collected and analysed at one specific point in time. This limits the research in understanding interactions and trends that occur in the framework over time. Nevertheless, backed-up with longitudinal studies on the same research area, links are able to be made to understand more about the development of the UWSS in Mexico City.

5.2 Interpretation of the results

The results of the CPA framework in the three dimensions convey a clear message about the current state of the UWSS in Mexico City. The capital values for availability, robustness and risk differ widely per capital. By analysing them integrally, it forms the foundation for the improvement of the UWSS in Mexico City in the future.

The availability of the Natural capital (N) is the only capital that is in abundance, meaning capital > 1.00 and the availability exceeds its demand. From this, we can tell that there is in fact enough water available, but that the problems of UWSS lie with the accessibility of these water resources to the public (Gutierrez, 2019). The low physical capital (P = 0.38) reflects the concerns in the literature, especially regarding leakage, land subsidence and aging infrastructure (Chen & Bilton, 2022). These three aspects have a strong reinforcing interaction with each other. High leakage rates cause more water to be pumped up to match demand, inciting more land subsidence, which in turn damages the aging infrastructure and increases the leakage rates. This is a negative feedback loop that is at the root of the Mexico City UWSS problem and has negative effects on the hydrology and geomorphology (Huerta-Vergara, Arciniega-Esparza, Pedrozo-Acuña, Matus-Kramer, & Vega-López, 2022; Machowski, Rzetala, Rzetala, & Solarski, 2016). The lack of physical capital impacts the financial capital (F = 0.43) in the ability to improve the current system. This is in line with the lack of funding made available for the system to function properly (Hernández, 2022). It demonstrates that the interconnection between P and F in the CPA framework reflects back well in reality. Also

the institutional capital (I = 0.50) is insufficient in its contribution to cover the public services. The agreements and mechanisms that manage water resources provide the partial presence of public services. The lack in institutional accountability is the main problem and is the large inhibitor for institutional public service coverage (Wilder, 2010, 2013). Especially the lack in executive accountability has been well established and identified as problematic in Mexico (SGI, 2022). The public services in Mexico City do not cover the public needs in water availability (CP = 0.54). Evidence for this is widely available, as inhabitants in several neighbourhoods are forced to resort to (1) household water harvesting, (2) purchase of bottled water, (3) illegal water tapping, or so-called *pipas*, (4) water rationing, (5) household purification of water (de Alba, 2017; Huberts et al., 2023; Lerner, Eakin, Tellman, Bausch, & Hernández Aguilar, 2018; Tortajada, 2006). These measures, amongst others, make up the adaptive capacity (A = 0.33) and helps households in the day-to-day survival to cover for the lack in public water services. Even though intraurban differences are large in Mexico City, the averaged needs for the dimension availability not met (CP+A = 0.87). This is also the reason why local bottom-up adaptation strategies, discontent in society and the need for sustainable water management persists in Mexico City (Eakin et al., 2020).

The robustness of the system shows similar focal points as the availability of the capitals. Natural capital robustness (N = 0.84) is covered to a large extent by the fact that there is little external dependency to obtain the required water resources, and that there is diversity in sources (Tortajada, 2006). The aspects in which N robustness is lacking, regard to the water quality and wastewater treatment. This is widely recognised throughout society, where 73% of the Mexican population relies on bottled water as primary source of drinking water (Greene, 2018). The rigidity of the infrastructure is a thorn in the eye of the UWSS. Maintenance is lacking, infrastructure is aging and water supply is discontinuous in many parts of the city (UNODC, 2021).

The red line of risk values shows the educated estimations for the risks to which the system is subjected. Some risks that have been a constant throughout the recent history and should be taken into account in the future as well, such as earthquakes and socioeconomic (in)stability (Alcántara-Ayala, Rodríguez-Velázquez, & Lucatello, 2022; Novelo-Casanova et al., 2022). There are also risks that shift throughout time and are likely to become increasingly more important in the future. For example, climate change will increase the risk of floods and droughts, but also land subsidence and unforeseen population growth are temporal

developments that are likely to pressurise the system. Adapting to temporal developments of risks by coming up with mitigation strategies helps in the effectivity of a swift response. The results show that there are perceived risks across all capitals. Due to the fact that individual risks can impact several capitals simultaneously, it is important to understand the dynamics behind the risks and how to combat these. By dealing with the root causes of the risks, the capital values of the risks will decrease consequentially.

Besides the abundance of available water, several other strengths have been identified in the system. These are the regulatory complexity, financial robustness and the federal city's storage capacity. The regulatory complexity covers the regional cooperation and intra- and interurban agreements. Mexico City has regulations and mechanisms in place to safeguard the water security in terms of regional water management disputes (Gobierno de la Ciudad de México, 2018). The independency of Mexico City and Mexico at large to cover their own production and demand for energy and water, instead of depending on other political entities, strengthens the financial robustness. Mexico produces 1.17 times more energy than it consumes, with natural gas, oil and hydrothermal energy sources making up for 79% of the total energy production (International Energy Agency, 2020). Lastly, the storage capacity (the recharge rates of the aquifers and the storage capacity of Cutzamala) gives large enough storage-to-flow ratio to serve as a buffer for unforeseen calamities. The storage-to-flow value is three times higher than the highest category of storage-to-flow in natural robustness.

There are also long-term trends that will affect the UWSS over time. Climate change in Mexico will lead to higher average surface temperatures (Ulises, Lucia, Delia, & Odila, 2022). This will in turn negatively impact the available water in the water cycle surrounding Mexico City. Especially with the necessity to shift from unsustainable groundwater extraction to more sustainable freshwater alternatives, higher evaporation of water and desertification of arable land leads to Mexico City becoming a less habitable place (Kamotho, Strahm, & Wolfangel, 2008).

With the prediction that Greater Mexico City will gain 2 million extra inhabitants throughout the upcoming three decades, on top of the current 22.3 million inhabitants, urbanisation will increase the pressures on the UWSS system (Hoornweg & Pope, 2017). More demand calls on the increasing necessity of usable water, when at the same time the mitigation of intracity socioeconomic differences has to lead to more equitable water

accessibility for all inhabitants. However, without proper financial and managerial interventions, trends in decreasing water quality, infrastructure and the consequential increase in social inequalities will follow (Mazari-Hiriart et al., 2019). These simultaneous processes can have cumulative negative impacts on the availability, robustness and the risks of the UWSS system. The results do show leverage points and weaknesses that come from the CPA analysis, which have practical and societal implications.

5.3 Practical and societal implications

Deriving from the results and its interpretations, it is evident that there are multiple implications to be made to the UWSS system at large in Mexico City. Noticing the abundance of water resources, but the lack of physical, financial and institutional capital, it tells us more about the necessary focal points. To deal with the bottlenecks in water security which are occurring and likely to aggravate in the upcoming decades, further investments and explorations for water sources is counterproductive (López-Morales & Mesa-Jurado, 2017). Misplaced focus on increasing the amounts of water resources draws away the attention from the actual bottlenecks. These are, amongst others, aged infrastructure, consequentially high leakage rates, ineffective and exclusive managerial strategies, and inadequate adaptive capacity to mitigate lacking public services.

The high leakage rates are, as aforementioned, the most important and influential aspect that can both mitigate and temper the negative effects of the UWSS in Mexico City. It limits the amount of land subsidence, due to more effective conversion of extracted water to supplied water (Ortiz-Zamora & Ortega-Guerrero, 2010). Replacing the aged infrastructure with newer, more flexible and less hazardous materials benefits the water quality, the robustness of the infrastructural system and the long-term cost-effectiveness (Palma et al., 2022). Investments in leakage control measures are most efficient in combatting the overexploitation of the aquifers, compared to investments into new water sources. Every dollar which is invested in leakage control measures has the same impact on the water supply as investing 8 dollars in searching for and pumping up new water sources (López-Morales & Mesa-Jurado, 2017).

Besides this leverage point, the system should also be wary for institutional weaknesses. Both *socio-economic stability* and *immediate threat to war or terrorism* are

influential in the effectiveness of the institutional capital. Risks regarding these two indicators have widespread consequences across Mexico, and especially its capital. By strengthening the institutional stability, the ability remains to govern more effectively, implement long-term interventions needed to safeguard the water supply security in Mexico City. Mexico has liabilities regarding migration from South America towards the USA, and faces internal conflicts with several drug cartels (Suarez, Villa, & Weiffen, 2017). If these trends are not dealt with accordingly and constructively, these pressures could rise to national level and impact the effectiveness of management, thereby impacting the clout in decision-making.

With challenges as complex as the UWSS in Mexico City, the societal implications of the results can be profound, depending on your socioeconomic background and location within the metropole. People living in less affluent neighbourhoods will have to keep on dealing with infrequent water deliveries, at non-potable quality standards and are dependent on personal adaptive skills. This raises political and ethical concerns. The current system does not support internationally agreed United Nations Sustainable Development Goal 6: Clean Water and Sanitation and the universal right to water (Spijkers, 2020). An encompassed strategy is needed that focuses on equitable distribution of clean water, sustainable management of non-renewable water sources and strengthens the system against avoidable and unavoidable risks.

5.4 Future research

Future research can focus on improving the methodology of the CPA framework, or its application to the case study of Mexico City. For the improvement of the CPA framework, extensive research on the interconnectivity between the capitals, dimensions and their indicators could prove to be fruitful. More scientifically proven associations between the indicators, e.g. the relationship between the decrease of socioeconomic stability and the adaptive capacity of households, can help standardise the quantification of the UWSS more accurately and increases its relevance for worldwide application.

In this case study, further research can delve into the exact effects of the leverage points, the weaknesses and potential unforeseen obstacles, such as effects of climate change that have not yet become problematic. These have to be overcome to ensure a full coverage of public water services. This can be further deepened by shock analyses, in which the

intensity of the robustness is tested and the effects of the risks are modelled. The goal of this extended research would be to test the rigidity and/or adaptability of the system to shocks in the future. By repeating this research over a timespan, the possibility is created to track the case-specific developments and give this research a temporal dimension. Overall, this research forms the basis for understanding the complexity of UWSS in urban cities worldwide. It is also the baseline from which future developments and in-depth research can start.

6. Conclusion

The case-study implementation of the CPA framework on Mexico City has given insight into the current state of the city's UWSS. Across the three different dimensions, there are a few overarching bottlenecks that cause a lack in public service coverage for the inhabitants of the metropole. The public services cover 0.54 of the necessary availability services and are compensated partially by social adaptation (+0.33). Robustness of the system is covered for 0.76 by public services, which adds up to 0.93 when social adaptation is included. This means that, including the social capital, there is insufficient water security and the system is not robust enough to withstand or recover from potential shocks to the system. Several chronic and acute risks, such as illegal tapping, earthquakes and floods are probable (average = 0.52; range across capitals 0.33-0.71) to occur in Mexico City. Preparing for the possible occurrence of these risks helps in mitigating the negative effects.

Prioritisation of leakage control, lack of managerial accountability and socioeconomic dissatisfaction and instability serve as the main focal points for the improvement of a sustainable urban water supply security. Tackling these problems would lead to more equitable distribution and availability of the water resources. Also, negative ecological and geomorphological pressures on the environment will decrease due to reduced water extraction, aquifer overexploitation and land subsidence. Managerial accountability will improve societal satisfaction, efficient governance and more inclusive decision-making.

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8. Appendix

CPA Framework Excel sheet: see file attached.